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Onsite MSBA Applied Project Report
Spring 2017
W.P. Carey, ASU

Topic	Data Driven Inventory Analysis (Compering preseason forecasts with InSeason realities)
Team	Team 110
Team Members	Sagar Pasad, Anwesha Banerjee, Ryan Karim, Zifan Wang, Kevin Jousha Marepally
Client information	SalonCentric (L'Oreal)

Executive Summary

Our project delivered a comprehensive, data-driven inventory optimization solution for SalonCentric, a subsidiary of L'Oréal USA that operates five distribution centers and over 600 retail stores serving more than one million stylists nationwide. The company's evolving digital transformation strategy required a more agile and scalable approach to managing its vast inventory across multiple brands and SKUs. Our goal was to bridge the gap between preseason forecasts and in-season actuals by integrating raw data, predictive modeling, and interactive visualizations to inform strategic and operational decisions.

We began by unifying disparate data sources, including Trend-3, Trend-5, and Trend-9 forecast projections, in-season actuals, Cost of Goods Sold (COGS), and SKU-level inventory details. A rigorous data pipeline was developed in Python to clean, standardize, and merge these datasets at the brand-month level. This created a consolidated analytical base for evaluating demand variability and inventory performance using financial and operational metrics such as Units, Dollars, and Days of Supply (DOS).

The core of our analytical approach involved implementing traditional inventory optimization metrics—Economic Order Quantity (EOQ), Reorder Point (ROP), and Safety Stock (SS)—with user-defined inputs like lead time, ordering cost, service level, and Mean Absolute Deviation (MAD). The resulting models enabled stakeholders to test what-if scenarios dynamically and explore the impact of different planning parameters on inventory health.

Interactive dashboards built in Tableau allowed cross-functional teams to visualize inventory risk levels, identify high-priority brands, and evaluate operational bottlenecks. Brands with low DOS (<15 days) were flagged as stockout risks, while those with DOS >30 days were identified as overstocked, incurring unnecessary holding costs. These insights supported smarter

replenishment strategies by transitioning from fixed reorder rules to flexible, data-informed decisions.

While the current solution focused on brand-month granularity for interpretability and performance, we acknowledged opportunities for deeper SKU-level and weekly analyses in the future. Data complexity, cost variability, and partial data gaps limited the full implementation of these refinements within the project timeline.

Looking ahead, we recommend integrating API-based data ingestion for near real-time updates, enhancing dashboards with additional filters (e.g., by region or product type), and implementing machine learning models for multi-objective optimization. Features such as anomaly detection for sudden DOS or cost deviations could further strengthen proactive decision-making.

In conclusion, our project empowered SalonCentric to move toward a more intelligent and scalable inventory planning framework, transforming siloed data into actionable insights. By aligning predictive models with business strategy, the solution enhances supply chain agility and supports sustainable, data-informed growth.

Background

SalonCentric, a subsidiary of L'Oréal USA, operates an extensive retail and distribution network across the United States, serving over one million professional stylists. With 622 stores and five distribution centers, the company manages a vast and diverse inventory across multiple brands and thousands of SKUs. As part of its broader digital innovation strategy, SalonCentric is undergoing a transformation in its supply chain operations—aiming to reduce distribution complexity, centralize purchasing, and adopt data-driven inventory practices.

Historically, the company relied on static forecasting models and manual inventory planning processes that lacked integration across systems. These approaches were insufficient for addressing real-time demand shifts, SKU-level variability, and operational constraints. As a result, the organization faced challenges such as stock imbalances, delayed replenishments, and limited visibility into actual inventory performance relative to preseason forecasts.

To support its strategic goals, SalonCentric required an automated and scalable solution capable of linking preseason planning models with in-season actuals while delivering actionable insights to cross-functional teams. This project was developed to meet that need by integrating historical data, predictive models, and interactive dashboards that enable proactive inventory decision-making.

Problem Statement

SalonCentric, with its expansive SKU base and multi-brand inventory structure, faced critical limitations in its existing inventory management system. The current tools lacked the flexibility and scalability needed to track thousands of SKUs across multiple brands, leading to challenges in maintaining supply-demand balance and reacting to in-season variability.

From a business perspective, this resulted in costly inefficiencies—stockouts affecting service levels and overstocking inflating holding costs. Strategically, there was no centralized system to align preseason forecast models (like Trend-3, 5, and 9) with in-season actuals across key dimensions like units, dollars, and Days of Supply (DOS). This hindered timely visibility into forecast accuracy, operational performance, and procurement agility.

Solving this problem was critical to enabling data-driven, real-time inventory planning. Demand planners, supply chain managers, and finance teams will use this solution to:

- Monitor brand performance monthly,
- Identify forecast deviations,
- Simulate what-if scenarios (e.g., lead time shifts or cost changes), ● Adjust planning strategies for improved service and efficiency.

We scoped the problem at the brand-month level to ensure analytical clarity and dashboard usability while maintaining scalability. Assumptions made included using uniform lead time and cost benchmarks where SKU-level data was unavailable and modeling demand variability using MAD as a proxy. This allowed for a unified, interactive solution that drives collaborative decision-making across teams.

Methods

Overall Approach

Our approach focused on creating an end-to-end, scalable inventory optimization system that could link preseason forecasts with in-season actuals and calculate key planning metrics such as Economic Order Quantity (EOQ), Reorder Point (ROP), and Safety Stock (SS). We followed a structured pipeline:

1. Data Integration: Merged siloed data sources across different planning horizons and inventory stages.

2. Data Cleaning and Transformation: Standardized, cleaned, and structured the data for analysis.
3. Analytical Modeling: Applied classical inventory formulas enhanced by user-defined variables (lead time, order cost, service level, MAD).
4. Scenario Simulation: Enabled interactive what-if testing to visualize the impact of different planning parameters.
5. Visualization: Delivered Tableau dashboards and Python visualizations for insight delivery.

We adopted this approach due to its balance of transparency, mathematical rigor, and practicality for non-technical business users. It allowed for rapid development while maintaining business interpretability.

Datasets Used

Collection & Extraction

We compiled data from historical inventory reports and demand planning [spreadsheets shared by SalonCentric](#). These included:

- Preseason forecast datasets based on Trend-3, Trend-5, Trend-9 models
- In-season actuals for Units, Dollar Value, and Days of Supply (DOS)
- Historical Cost of Goods Sold (COGS) data
- Product attribute files with SKU-level metadata

Sampling included monthly data across 12 months and covered 20+ brands and thousands of SKUs.

Nuances & Handling

- Missing Data: Some records lacked COGS or actuals for specific months. We retained such rows using outer joins, highlighting gaps instead of dropping them.
- Inconsistent Naming: Time fields (e.g., **Month**, **MONTH**, **Period**) were standardized.
- Duplicates were removed based on key dimensions (**Brand**, **Month**, **SKU**).
- Data was aggregated to the brand-month level to maintain performance and clarity in dashboarding.

Cleaning & Transformation

- Used Pandas for merging, cleaning, and standardizing.
- Normalized unit cost formats and filled minor missing numeric values with brand-level averages where needed.
- Created derived fields: **Weekly_Demand**, **Z-score**, **Safety Stock**, and **ROP**.

Analytical Models Used

We used deterministic inventory planning models based on classical operations research techniques:

Formulas

$$EOQ = \sqrt{DS/C}$$

$$ROP = \text{Lead Time} \times \text{Weekly Demand} + \text{Safety Stock}$$

$$\text{Safety Stock} = Z \times (1.25 \times \text{MAD}) \sqrt{\text{Lead Time}}$$

Why These Models?

- Well-established in supply chain literature
- Easy to implement, explain, and scale
- Effective under data limitations (no need for labeled training data)

Model Parameters

- User-defined: Ordering Cost, Lead Time (weeks), Service Level, MAD
- Derived: Weekly Demand (from forecast), Unit Cost (from COGS), Z-score (from service level)

Features

- Categorical: Brand, SKU, Month
- Numeric: Forecast Qty, Unit Cost, COGS, DOS, MAD

Training, Testing, Validation

As these are formula-based models (not machine learning), there was no training/validation split. However, validation was done through:

- Manual checks across brands for accuracy
- Testing outputs against known business rules
- Peer reviews with domain experts

Model Performance

- Outputs aligned with known replenishment behaviors
- Outliers (e.g., extremely high ROPs) were successfully flagged as risk cases
- Stakeholder feedback validated the usefulness of reorder recommendations

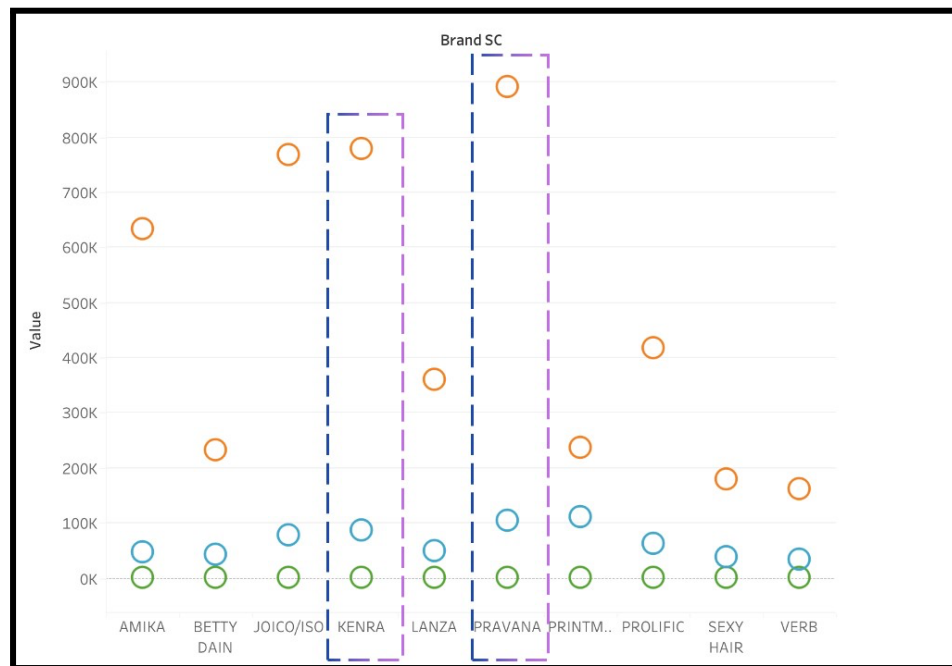
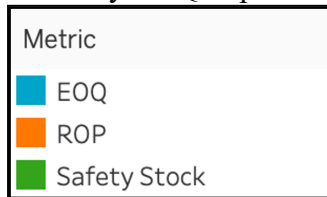
Tools & Platforms Used

- Python (Pandas, NumPy, Matplotlib, SciPy): Data processing, modeling, and plotting
- Tableau: Dashboard creation and stakeholder presentation
- Excel: Initial data formatting and quick checks
- Google Drive/Sheets: Collaborative data review and versioning

Results and Conclusions

Key Results

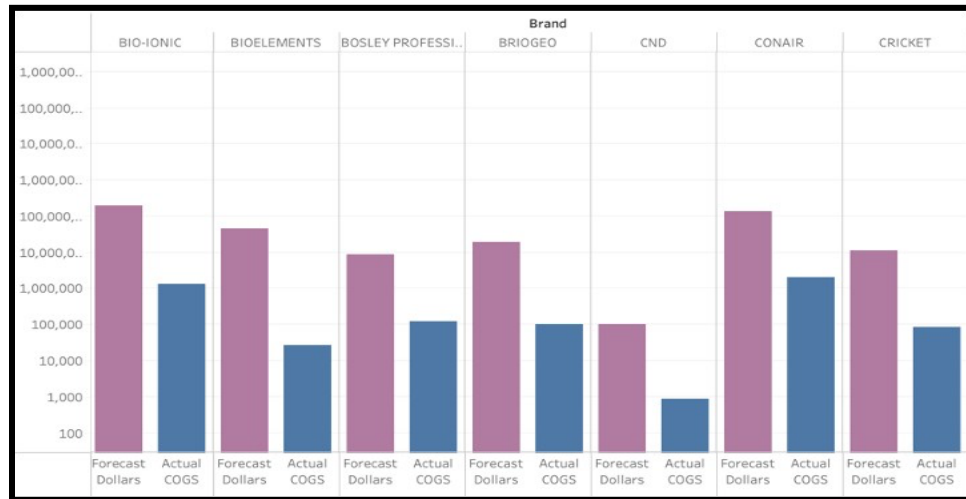
- We developed a scenario-based inventory optimization model that calculates Economic Order Quantity (EOQ), Reorder Point (ROP), and Safety Stock (SS) for multiple brands using user-defined parameters (Order Cost, Lead Time, Service Level, and MAD)
- The solution allowed real-time scenario testing through interactive inputs and ranked brands by EOQ to prioritize supply chain planning.



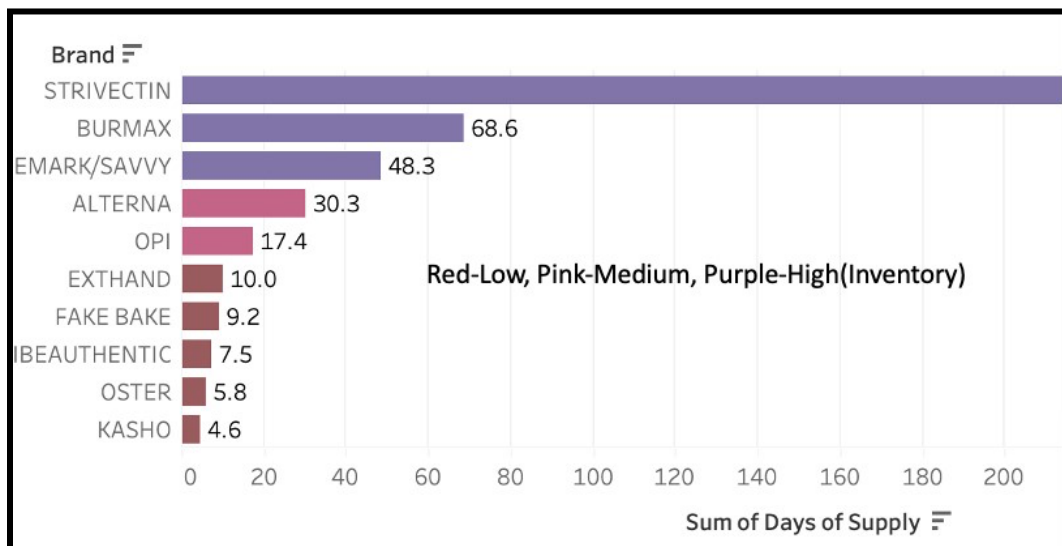
Real Time Scenario of the data provided

- Our analysis exposed mismatches between forecasted units and actual inventory, highlighting operational inefficiencies such as procurement delays, supplier issues, and inventory overstock or shortages.

Actual VS Predicted Mismatched Data



- Color-coded dashboards helped identify inventory health by categorizing brands as:
 - Red: Risk of stockout (DOS < 15 days)
 - Pink: Medium inventory (healthy range)
 - Purple:** Overstock risk (DOS > 30 days)



Business Impact

- Provide demand planners and supply chain managers with a centralized, data-driven tool for timely replenishment decisions.
- Help reduce stockouts, lost sales, and excess holding costs.
- Improve the visibility and reliability of inventory planning by comparing forecasted versus actual performance across brands and months.
- Enable cross-functional collaboration by giving finance, planning, and operations teams a shared analytical view of inventory health.

Users & Usage

The solution is designed for:

- Demand Planners: To evaluate forecast accuracy and plan replenishment dynamically.
- Supply Chain Managers: To adjust procurement and distribution decisions based on lead time sensitivity and risk zones.
- Finance Teams: To track inventory dollars and align with budget expectations. Executive Leaders: To assess operational performance and resource allocation through strategic dashboards.

References

1. [SCM 502 Certificate Module 4 Inventory Study Guide](#)
2. [SCM 502 Certificate Module 2 Forecasting Study Guide -1](#)
3. SciPy Documentation – `scipy.stats.norm.ppf`
<https://docs.scipy.org/doc/scipy/reference/generated/scipy.stats.norm.html>
– Used to calculate Z-scores for service level targets in safety stock computations.
4. Pandas Documentation – Data manipulation and merging <https://pandas.pydata.org/docs/>
– Used extensively for data cleaning, transformation, and outer joins.
5. Tableau Public Documentation <https://help.tableau.com/current/pro/desktop/en-us/help.htm>
– Used to create interactive dashboards for visualizing inventory trends.
6. Google Colab Notebooks <https://colab.research.google.com>
– Used as the development environment for the Python-based modeling pipeline.
7. Fair Use Code Examples from Stack Overflow (Accessed March–April 2025)
<https://stackoverflow.com>
– Referenced for Python syntax and troubleshooting during data processing and visualization.
8. SalonCentric (L’Oréal USA) internal project documentation and shared datasets – Used with permission under academic fair use for analysis and visualization purposes. Brand names were masked as required for confidentiality.

Appendix – Reproduction of Results

1. Data Sources

The following datasets were used for this analysis:

Dataset Name	Description	Access Info
forecast_trend_3_5_9.csv	Preseason forecast data using Trend-3/5/9 models	Provided by sponsor (SalonCentric)
inventory_actuals.csv	In-season inventory units and DOS per brand/month	Provided by sponsor
actual_COGS.csv	In-season cost of goods sold	Provided by sponsor
sku_level_inventory.csv	SKU-level historical data with MAP cost	Provided by sponsor
product_attributes.csv	Brand/SKU metadata	Provided by sponsor

Note: For confidentiality, masked datasets with anonymized brand names have been submitted on Blackboard under the "Data Files" folder.

2. Tools and Platforms

Tool/Platform	Purpose	Access Information
Python (v3.10)	Data cleaning, modeling, scenario analysis	Use Anaconda or Google Colab
Google Colab	Cloud environment for running Python	Link to Notebook
Tableau (2023.1)	Dashboard creation and visualization	Tableau Desktop (student license) or Tableau Public

Excel	Manual data verification, CSV reviews	Local or Google Sheets
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3. Scripts Used

All scripts have been submitted on Blackboard under the folder:

[Capstone_Team110 > Scripts and Notebooks](#)

Script Name	Purpose
01_data_merge_and_clean.py	Merges forecast, actuals, and COGS datasets
02_eoq_rop_safetystock.py	Computes EOQ, ROP, and Safety Stock using user inputs
03_visualization_output.py	Generates bar and line plots for top SKUs
dashboard_data_output.csv	Output CSV used for Tableau dashboard creation

4. How to Run the Scripts

Option A: Using Google Colab

1. Open the Colab link: [Inventory Optimization Notebook](#)
2. Upload your cleaned [.csv](#) files to Colab
3. Run cells sequentially — interactive sliders will allow you to simulate EOQ, ROP, SS values
4. Output will be saved as [calculated_inventory_forecast_metrics](#)

Option B: Local Python (e.g., Anaconda)

1. Download all [.py](#) files and datasets from Blackboard
2. Set your working directory to the folder with the data

Run scripts in order:

[python 01_data_merge_and_clean.py](#)

[python 02_eoq_rop_safetystock.py](#)